

An integrated population-averaged approach to the design, analysis and sample size determination of cluster-unit trials

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SUMMARY

While the mixed model approach to cluster randomization trials is relatively well developed, there has been less attention given to the design and analysis of population-averaged models for randomized and non-randomized cluster trials. We provide novel implementations of familiar methods to meet these needs. A design strategy that selects matching control communities based upon propensity scores, a statistical analysis plan for dichotomous outcomes based upon generalized estimating equations (GEE) with a design-based working correlation matrix, and new sample size formulae are applied to a large non-randomized study to reduce underage drinking. The statistical power calculations, based upon Wald tests for summary statistics, are special cases of a general power method for GEE. Copyright © 2003 John Wiley & Sons, Ltd.

KEY WORDS: cluster randomization; generalized estimating equations; logistic models; power

1. INTRODUCTION

1.1. Motivation and literature review

Population-averaged (PA) models provide a natural approach for the analysis of cluster-unit trials since differences in population means or proportions between intervention and control conditions are typical parameters of interest. While the statistical literature has emphasized a fully parametric mixed model approach to cluster-unit trials, less attention has been given to analysis using PA models. This paper describes a novel implementation of the generalized

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estimating equations (GEE) approach based upon simple specifications of the design-based correlation structure. These methods are applicable to many government and foundation-sponsored programmes that provide support for cluster-unit trials enrolling a moderate to large number of clusters [1]. Although this paper focuses on large cluster-unit trials, recent advances in GEE methods (see discussion) suggest the methods presented here have wide applicability.

Another area receiving scant attention in the literature is the design and analysis of group non-randomized trials. Randomization is often unobtainable in many government and foundation-sponsored programmes. Non-randomized trials with assignment of interventions to intact groups of individuals create special difficulties in sample selection. A consequence of group non-randomized trials is that communities selected to receive intervention may differ from those not selected with respect to important characteristics. Those differences often lead to biased estimates of the intervention effect. The propensity score, defined as the conditional probability of receiving the intervention given covariates, can be used to balance the covariates in the two groups, and therefore reduce this bias [2]. Conversely, traditional methods of adjustment – matching, stratification and covariance analysis – may be inadequate; matching and stratification are restricted to categorical variables and traditional analysis of covariance depends upon the correct function of the covariables in the statistical model [3]. All three methods are limited in the number of covariates for adjustment, whereas propensity scores, which provide a scalar summary of the covariate information, can deal with a very large number of covariates. This paper presents a design strategy that uses propensity scores to select matching control communities to manage the selection bias in a large cluster-unit trial.

Experimental designs have a long history with random assignment playing a notable role for assessing the effect of social interventions [4, 5]. More recently, both group randomized and non-randomized trials have been increasingly used in public health research. Other names for such studies are ‘community trials’ and ‘cluster-unit trials’. This article makes no important distinction between the use of the various terms. A primary feature of cluster-unit trials is the assignment and implementation of an intervention at the group or community level, while measurements are made on subjects sampled within groups. An implication of the assignment of interventions to groups of subjects is that correlation among subjects within a group needs to be accounted for in the planning and conduct of the analysis. Failure to do so may result in inflated type I error, leading to invalid conclusions [6].

Numerous statistical approaches have been proposed for the analysis of cluster-unit trials, including an integrated mixed model approach to the design, analysis and sample size determination of group randomized trials [6–8]. Other work has focused on randomization tests applicable to trials that randomize a small number of communities [9, 10]. Recently, statistical analysis methods for both subject-specific (mixed) models and PA models for dichotomous outcomes were reviewed [11, 12]. There is less consensus for the proper analysis of categorical outcomes than for continuous outcomes in cluster-unit trials [13]. We present an integrated PA model approach, based upon GEE, to the design, analysis and statistical power of group intervention trials encompassing the analysis of both discrete and continuous outcomes but with a special focus on the former. Our methods for statistical analysis and power are applicable to both group randomized and non-randomized trials, whereas our discussion of design issues considers the special difficulties of designing valid and cost-effective non-randomized community trials.

Two distinct analysis strategies are available to adjust for the effect of clustering using GEE [13]. The approach that has received primary consideration in the cluster-unit trials literature is based upon empirical sandwich or so-called 'robust' covariance estimators [11, 12]. The advantage of this method is that valid inference is obtained in an asymptotic sense even if the working correlation structure has been misspecified. In this approach, simple 'working correlation' matrices are specified, such as independence and exchangeable structures, even though these may not accurately describe the pattern of correlation between responses of cluster members. One important exception is the post-test-only nested cross-sectional design, in which the exchangeable correlation may provide an adequate description of the intracluster correlation structure. Obtaining valid inference using the GEE approach based upon empirical sandwich covariance estimation requires approximately 40 clusters, as shown by several simulation studies [11, 14]. The poor performance of the empirical sandwich estimator with fewer clusters has also been shown in the linear mixed model setting [15]. In view of such results, the use of GEE methods has traditionally been thought to be not applicable to studies with fewer than 20 clusters per intervention condition [7]. Although this paper does not address the performance of estimating equation methods when there are a small number of clusters, recent research into the performance of GEE with a bias-corrected covariance estimator suggests the scope of GEE methods for cluster-unit trials may be broader than what has been shown previously [14].

The second strategy, and the one adopted here, is the model-based approach. It requires the specification of a design-based working correlation matrix which describes the pattern of correlation between responses of cluster members [13]. Whereas the model-based variance estimator has received scant consideration in the group randomized trials literature, an analogous model-based approach is routinely adopted for maximum likelihood-based mixed models, whereby the correlation structure is determined through specification of random effects [7, 8, 11]. Three advantages accrue from a strategy based upon model-based variance estimators. First, careful modelling of the correlation structure may increase statistical efficiency of the GEE regression parameter estimates [16]. Second, even for large cluster-unit trials, the model-based variance estimator may have performance superior to the robust variance estimator. In a simulation study based upon 193 clusters each with a cluster size of 88, the GEE empirical sandwich variance estimator was found to be highly variable compared to a correctly chosen model-based estimator based upon a spatial correlation structure [17]. Third, as in the variance components approach, a benefit of the model-based approach is that estimates of intraclass correlations can be used in the power analyses of future studies [18–21].

Once the design and statistical analysis plan have been identified, an appropriate power analysis is carried out to determine sample size. We propose sample size formulae that require explicit specification of intraclass correlations from PA models. The idea is to use traditional and easy-to-apply power analysis methods for independent samples treating a summary community-level statistic as the unit of analysis. The main task for a particular study design is to define the summary statistic and determine its variance in terms of the different types of intracluster correlations. A sample size formula applicable to discrete and continuous outcomes is derived for the pre-test–post-test nested cross-sectional design. The proposed formula is a special case of a general method for power based upon GEE [22]. It is also related to sample size formula applicable to mixed models [8]. This paper presents an integrated PA model approach with sample selection using propensity scores, statistical analysis based upon

GEE, and statistical power based upon summary statistics. The methods are illustrated with binary data from the national evaluation of the Enforcing Underage Drinking Laws Program, a nationwide non-randomized community trial to reduce underage drinking, which we simply refer to as the national evaluation.

1.2. An example: reducing underage drinking

The Enforcing Underage Drinking Laws Program, launched by the United States Office of Juvenile Justice and Delinquency Prevention (OJJDP) in 1998, is the largest federal initiative focused on reducing underage drinking in U.S. history. Each of the 50 states and the District of Columbia receives significant funding and technical assistance to support state and local efforts to enforce laws related to alcohol use by underage persons and to prevent underage drinking. A major component of the programme involves discretionary grants provided to a subset of the states on a competitive basis. In the first year of the programme, discretionary grants were awarded to nine states. These states awarded subgrants to 61 communities according to criteria which are not entirely known to the national evaluation team and which may vary across states. An additional intervention community from one of the states (Minnesota) could not be part of the evaluation due to a missing code which precluded linkage to census data. A tenth state that awarded subgrants to a different set of communities other than what was initially specified was also not part of the evaluation. With respect to the nine states, randomization was not employed and the evaluation team (led by the fourth author) had no control over the choice of intervention communities. The selection of control communities by the team is described in Section 2.

The evaluation of the impact of the discretionary grants is carried out with a nested cross-sectional design. Data are collected using a variety of surveys; in this paper we focus on repeated telephone surveys of 15 to 20 youths aged 16–20 years in each selected community. The outcomes of interest are self-reported past 30-day alcohol use and binge drinking, defined as consumption of five or more drinks in one drinking occasion. Data collection was conducted pre-intervention (or early in the intervention period), one year later, and two years later. This paper considers the change in past 30-day alcohol use from baseline to one year assuming a pre-test–post-test nested cross-sectional design [7]. A cross-sectional design was chosen over a cohort design since the long intervention period might result in substantial drop-out in a cohort design [8, 13]. Additionally, our interest focused on the change in the population over time as opposed to within-individual change.

1.3. Organization of the paper

Sections 2, 3 and 4 present design, analysis and sample size issues, respectively, for cluster-unit trials. Design issues in Section 2 concern the special challenges arising in the planning of large non-randomized trials. In particular, the propensity score method as employed in the design of the national evaluation is described. In Sections 3 and 4, a description of the general approach to statistical analysis and sample size determination is followed by application to the national evaluation. Section 5 presents a summary and discusses the applicability of the proposed methods for cluster-unit trials with a small to moderate number of communities.

2. DESIGN CONSIDERATIONS FOR NON-RANDOMIZED CLUSTER-UNIT TRIALS

2.1. Use of propensity scores to select control communities in the design of a large non-randomized community trial

The first major task of the national evaluation team was to identify a sampling scheme that would select a subset of control communities to provide matches to all or most of the 61 grant communities. Based upon power considerations (as presented in Section 4.5), a goal of 54 matches was established; 52 matches were obtained using a propensity score matching method applied individually to states. States include those (Connecticut, Louisiana, and Pennsylvania) with communities (that is, the units of intervention) defined as cities and the six states (Maryland, Michigan, Minnesota, New Mexico, Ohio and Wisconsin) where counties are sampled. The numbers of grants awarded by states are shown in Table I along with the number of available prospective control communities. This latter group included all communities within a state not receiving intervention with population at least as large as that of the least populated intervention community. Since a study including all prospective communities would be too costly and generally not feasible, the goal of the national evaluation team was to select intervention and control sites by forming matched pairs based upon similar characteristics as captured by propensity scores. The propensity score is a scalar summary of several community characteristics captured by the U.S. census or other external data sources and identified *a priori* as being likely related to the grantee selection process and the major outcomes to be measured by the youth survey.

The first step in the selection process is to fit a logistic regression model to a state's full set of communities using intervention status as the dependent variable. We used four community characteristics, population size, median income, number of liquor law arrests per 100000 population [23] and size of college population (adjusted for overall population) [24]. For states where communities were defined as counties, 1998 population [25] and 1993 median income [26] were used. For states where communities were defined as cities, 1990 population and median income were used [27]. The logistic regression model gives a predicted probability of being an intervention community, which is called a propensity score. The second step was to order intervention communities according to their propensity score from largest to smallest. For each one, a set of potential controls was identified as those having an absolute difference in propensity score from the intervention community of less than 0.05. We refer to 0.05 as the 'caliper', as it indicates an acceptable distance between the sets of covariate values for an intervention community and its potential matched control. A control community was selected if it was the only one in the set. If the set contained more than one control, a random selection process was used. We ordered the controls within a set according to preference for selection. Highest preference was given to control counties (or cities) that were not geographically adjacent to intervention counties and that did not receive any small grants under the national evaluation block grant programme. The presence of either of these conditions might adversely affect estimation of the intervention effect through contamination of the control condition. Least preference was given to control counties that were adjacent to intervention counties and had small grants. Control counties with exactly one of these characteristics constituted a group, which we call the 'middle' group, that received second highest preference. A control county was randomly

Table I. Number of communities in population and sample by state.

State	Population		Sample			
	Prospective intervention communities	Prospective control communities	Matched community pairs	Adjacent control communities	Small grants	
					Intervention communities	Control communities
CT	6	17	3	3	0	0
LA	3	34	3	0	1	1
MD	5	18	5	1	0	4
MI	11	72	9	0	0	0
MN	13	74	12	1	0	0
NM	7	26	5	2	0	0
OH	4	84	3	0	0	0
PA	5	99	5	0	1	0
WI	7	65	7	0	0	0
Total	61	489	52	7	2	5

selected from the group of highest preference. If the set for highest preference was empty, the control was selected from the middle group. A control was selected at random from the least preferred group only if there were no sites in the highest or middle group. Finally, some intervention communities had no matches and, with few exceptions, were not included in the data collection and analysis. Once a control was selected it was removed from the available pool of controls for subsequent intervention communities. In summary, the goal of the propensity score algorithm is not to achieve one-to-one matching of communities, but rather to achieve balance, on average, with respect to important community characteristics.

2.2. Results of the community selection procedure

The propensity score algorithm resulted in 48 matches. Modification of the algorithm was employed for some states, and, in a few instances, additional criteria were used to increase the total number of matches to 52. States with cities serving as communities (CT, LA, PA) received special consideration. For Louisiana and Pennsylvania, the algorithm was applied after first excluding from possible selection any prospective control city within a county of an intervention city. For Connecticut, all candidate control cities, but one, were within counties with intervention cities, so this restriction was not made, although preference was given to the one exception. In CT and LA, the city selection algorithm with a caliper of 0.05 failed to identify the requisite minimum number of three matches established by the evaluation team for each state. One source of difficulty with LA is that there was a very limited number of possible matches because there were only three intervention cities. For Connecticut, the limitation was due to relatively few potential controls (Table I). In order to achieve at least three matches per state, one match in CT and two matches in LA were made by selecting the control with the propensity score nearest the intervention cities but outside the 0.05 caliper. In PA, a caliper of 0.005 was used to obtain three matches since 0.05 would have given a larger number of potential matches.

Table II. Mean population size in thousands (standard deviation) before and after matching.

State	Before matching		After matching	
	Intervention communities ($n_1 = 61$)	Control communities ($n_2 = 489$)	Intervention communities ($n_1 = 52$)	Control communities ($n_2 = 52$)
CT	67 (25.0)	72 (35.0)	59 (14.0)	55 (9.0)
LA	13 (4.0)	48 (94.0)	13 (4.0)	14 (6.0)
MD	194 (298.0)	264 (287.0)	194 (298.0)	179 (331.0)
MI	49 (41.0)	128 (299.0)	53 (44.0)	42 (39.0)
MN	139 (290.0)	38 (58.0)	68 (132.0)	45 (92.0)
NM	59 (52.0)	51 (102.0)	64 (61.0)	36 (43.0)
OH	278 (248.0)	120 (211.0)	351 (247.0)	340 (445.0)
PA	44 (46.0)	35 (158.0)	44 (46.0)	28 (19.0)
WI	87 (59.0)	70 (127.0)	87 (59.0)	62 (32.0)

Modifications to the algorithm were made for Maryland and Minnesota. In Maryland, all controls in the pool but one received small grants and most were adjacent to an intervention county. Applying the algorithm with preferences as defined above resulted in poor matches since the possibilities were greatly diminished. A modification of the algorithm for Maryland was made by combining the middle and least preferred groups. Lastly, in Minnesota, the three intervention cities with the largest population (and largest propensity scores) were not selected by the algorithm. To increase representation we added a match of one of these large cities to the control judged most similar with respect to the four characteristics.

Table I summarizes the 550 communities in the population and the 104 communities selected into the sample. Only seven controls were selected that were adjacent to intervention communities. Also, the number of communities with small grants chosen was low; Maryland was an exception where four out of five control communities received a small grant. From a pool of 489 prospects, 52 controls were identified. Those intervention communities left out were the ones that were hardest to match with respect to the four characteristics.

The effectiveness of the selection process is evaluated by comparing means of the characteristics for control and intervention communities before and after selection. The difference in means between controls and intervention communities for the covariates population size (shown in Table II), median income and liquor law violations (not shown) was reduced by the sample selection process for six out of nine states. The difference in means between controls and intervention communities for the covariate college attendance (not shown) was reduced by the sample selection process for seven out of nine states. These results were consistent with giving the four covariates equal consideration in the selection algorithm. If results are collapsed over states, the difference between means is reduced for only one of four characteristics. Since the propensity score algorithm was applied to each state, this result is not surprising, but supports use of a statistical model to test the intervention that includes propensity score as a covariate or state as a factor.

Results from the pre-test period reveal that the propensity score algorithm achieved the desired balance. Pre-intervention results on 1749 youth from 104 communities revealed past

30-day alcohol use at 43.4 per cent and 43.2 per cent for intervention and control communities, respectively. A comprehensive report on the pre-intervention results is available.

The propensity score method is particularly useful for the design of non-randomized studies when there are several community-level characteristics for which balance across conditions is desired. In the national evaluation, four community characteristics known *a priori* through U.S. census or other publicly available data were used for matching. Subsequent to the initial planning phase of this project, additional funds were provided to add 17 new intervention and 17 new control communities (from additional states not considered here) at the second and third timepoints. In order to extend the national evaluation to this second wave of funding of discretionary grants, the propensity score method was applied again with some modification of the algorithm described here taking additional characteristics into consideration based upon pre-intervention results [28].

3. A POPULATION-AVERAGED MODEL APPROACH TO THE ANALYSIS OF CLUSTER-UNIT TRIALS

3.1. Generalized estimating equations

We describe GEEs for the pre-test–post-test nested cross-sectional design of the national evaluation. The proposed methods can be easily extended to cohort designs and/or designs with multiple time points. Assuming there are K sampled communities, let $Y_i = (Y_{i0}, Y_{i1})'$ be the response vector for community $i = 1, \dots, K$ of dimension $m_{i0} + m_{i1}$ where $Y_{it} = (Y_{i1t}, \dots, Y_{im_{it}t}, \dots, Y_{im_{it}t})$ is the vector of responses from the m_{it} individuals in the i th community at time $t = 0$ (pre-test) or $t = 1$ (post-test). Let π_i be the vector of population marginal means such that $E[Y_i] = \pi_i$. A generalized linear model is $g(\pi_{ijt}) = X'_{ijt}\beta$ where $g(\cdot)$ is the link function and X_{ijt} is the $p \times 1$ vector of covariates for the j th individual in the i th community at time t . Estimation of β is performed by solving

$$\sum_{i=1}^K D'_i(X_i, \beta) V_i^{-1} [Y_i - \pi_i(\beta)] = 0 \quad (1)$$

where $D_i(X_i, \beta) = \partial \pi_i / \partial \beta$, $V_i = A_i R_i A_i$, where $A_i = \text{diag}\{v_{ijt}^{1/2}\}$ is a diagonal matrix with $v_{ijt} = v(\pi_{ijt}) = \text{var}(Y_{ijt})$ and $R_i(\rho)$ is a working correlation matrix that depends on an unknown nuisance parameter vector ρ . Estimation of β is by iteratively reweighted least squares alternating with method of moments estimation of ρ [16]. The GEE estimate $\hat{\beta}$ has an asymptotically Gaussian distribution with mean β and covariance matrix estimated by $(\sum_{i=1}^K D'_i V_i^{-1} D_i)^{-1}$, the model-based variance estimator which provides valid inference when R_i is correctly specified (that is, $\text{corr}(Y_i) = R_i$). Two intraclass correlations are specified, $\rho = (\alpha_0, \alpha_1)$ where $\alpha_0 = \text{corr}(Y_{ijt}, Y_{ikt})$, $\alpha_1 = \text{corr}(Y_{ijt}, Y_{iks})$, for $j \neq k, s \neq t$. These are the correlation between outcomes from different youths at the same time (α_0), and the correlation between two outcomes from different youths at different times (α_1). Then

$$R_i(\alpha_0, \alpha_1) = \begin{pmatrix} R_{i1} & R_{i12} \\ R_{i12}^T & R_{i2} \end{pmatrix}$$

where $R_{it} = (1 - \alpha_0)I_{m_{it}} + \alpha_0 J_{m_{it}, m_{it}}$, $R_{i12} = \alpha_1 J_{m_{i1}, m_{i2}}$, I_r is the identity matrix of dimension r , and $J_{r,s}$ is the $r \times s$ matrix of 1s. For a nested cohort design, there is an additional correlation, $\alpha_2 = \text{corr}(Y_{ijt}, Y_{ijs})$, describing the association among responses from two time points on the same individual. It replaces certain α_1 elements in R_{i12} . We use Liang and Zeger's all-available-pairs estimator for each element of ρ [16, 29]. SAS/IML software [30] to fit GEE with this special correlation structure is available from the authors.

3.2. Statistical analysis for the national evaluation

Differences in changes in self-reported past 30-day alcohol use ($Y_{ijt} = 1$ if yes, 0 if no) are evaluated with GEE. The hypothesis that control and intervention practices differ in the change in prevalence over time may be tested by fitting a generalized linear model

$$g(\pi_{ijt}) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2t} + \beta_{12} x_{1i} x_{2t} \quad (2)$$

where $\pi_{ijt} = E(Y_{ijt}) = P(Y_{ijt} = 1)$ is the probability of past 30-day use for the j th youth in the i th community at time t in the intervention ($x_{1i} = 1$) or control group ($x_{1i} = 0$) at either pre-test ($x_{2t} = 0$) or post-test ($x_{2t} = 1$). Note that β_0 corresponds to the prevalence at baseline for control communities, β_1 represents the difference in prevalence between intervention and control communities at pre-test, and, since $x_{2t} = t$, β_2 represents the increase in prevalence for control communities from pre-test to post-test. The intervention effect is given by β_{12} , which is related to the change in prevalence over time between control and intervention practices. For the logit link, $g(\pi_{ijt}) = \log(\pi_{ijt}/(1 - \pi_{ijt}))$, β_{12} is the contrast between two log-odds ratios: the post-test log-odds ratio of drinking among the intervention group communities versus the control group communities minus the analogous pre-test log-odds ratio. Sample selection using propensity scores partly addresses the imbalances among covariates that may exist between intervention and control communities. Model equation (2) may be expanded to include different intercepts for the nine states, a covariate containing the propensity scores or important demographic variables such as gender and age in order to address any post-sample selection imbalances among covariates. Finally, in equation (1), $v_{ijt} = \pi_{ijt}(1 - \pi_{ijt})$ and the design-based correlation matrix is $R_i(\alpha_0, \alpha_1)$.

Summary proportions of past 30-day alcohol use in the national evaluation are reported in Table III. Overall, the percentage of past 30-day alcohol use decreased by 3 per cent among youth in control communities and by 5 per cent among youth in intervention communities. Pre-test proportions and changes in proportions from pre-test to post-test varied considerably by state and age.

Youths of 19 and 20 years of age were more likely to drink than youths 18 and younger, but they were also less likely to be sampled. The GEE results for three marginal models, each estimated using three different correlation structures, are shown in Table IV. Each model fit is based upon $N = 3661$ observations, and an average cluster size of 35.2, or 17.6 observations per community and time point. Although the interpretation of the intervention effect given by the intervention by time interaction in the three models depends upon whether or not additional variables are conditioned upon, all results show a non-significant intervention effect. Since $\hat{\beta}_1$ ('Intvn' in Table IV) is approximately zero, the parameter, $\exp(\beta_{12})$,

Table III. Rates of alcohol use in past 30-days (and number of respondents, N) for the national evaluation of the Enforcing Underage Drinking Laws Program.

	Control communities		Intervention communities	
	Pre-test % (N)	Post-test % (N)	Pre-test % (N)	Post-test % (N)
Overall	43 (898)	40 (942)	43 (877)	38 (944)
State				
CT	42 (57)	37 (60)	44 (50)	38 (60)
LA	52 (56)	39 (54)	49 (57)	44 (48)
MD	54 (84)	38 (82)	59 (80)	38 (87)
MI	47 (161)	43 (170)	44 (154)	39 (158)
MN	45 (196)	48 (218)	39 (205)	40 (227)
NM	37 (97)	32 (94)	40 (81)	32 (88)
OH	31 (52)	33 (55)	29 (52)	29 (52)
PA	28 (85)	28 (88)	36 (81)	28 (80)
WI	45 (110)	41 (121)	52 (117)	46 (144)
Gender				
male	46 (452)	36 (470)	44 (432)	35 (454)
female	40 (446)	44 (472)	43 (445)	41 (490)
Age, years				
16	34 (257)	27 (293)	29 (231)	26 (304)
17	38 (250)	38 (292)	38 (272)	31 (294)
18	47 (189)	48 (167)	46 (176)	53 (174)
19	55 (124)	52 (107)	65 (118)	59 (95)
20	64 (78)	60 (83)	66 (80)	55 (77)

which is a ratio of odds ratios, may be interpreted, for practical purposes with respect to the national evaluation, as a post-test odds ratio. For example, for the GEE with design-based correlation structure, the model 3 result of $\exp(\hat{\beta}_{12}) = \exp(-0.094) = 0.91$ means that a youth in an intervention community has about 0.91 times the odds of drinking in the past 30-days as a youth in a control community at post-test. Results of model 3 show a statistically significant age effect and a marginally significant effect due to propensity scores after adjusting for between-state variation through specification of state as a block effect by allowing different intercepts. Finally, there is evidence that the design-based correlation structure is a more adequate description of the true correlation structure than independence or exchangeable assumptions. First, the estimate of $\hat{\alpha}_0 = 0.021$ differs from $\hat{\alpha}_1 = 0.007$. Next, the model-based and robust standard errors for the intervention effect are very similar under the design-based correlation structure. Although the GEE method as implemented here does not provide standard error estimates for α , we can assume these estimates are reasonably precise since they are based upon 104 communities. As such they can be used for planning future community interventions with the outcome past 30-day alcohol use. The estimating equations method of Zhao and Prentice provides standard error estimates of α [31].

Table IV. GEE logistic model parameter estimates and model-based standard errors (empirical standard errors) for past 30-day alcohol use.

Working	Variable	Model 1		Model 2		Model 3	
correlation		Estimate	S.E.	Estimate	S.E.	Estimate	S.E.
Independence	Intvn	0.002	0.096 (0.113)	−0.005	0.097 (0.112)	0.005	0.099 (0.120)
	Time(post)	−0.144	0.095 (0.090)	−0.148	0.095 (0.090)	−0.104	0.098 (0.096)
	Intvn*time	−0.085	0.135 (0.148)	−0.082	0.136 (0.149)	−0.083	0.139 (0.155)
	prop.score			0.641	0.328 (0.380)	0.735	0.337 (0.405)
	male gender					0.217	0.070 (0.068)
	age = 18					0.689	0.089 (0.087)
	age = 19-20					1.073	0.088 (0.087)
Exchangeable	Intvn	0.005	0.105 (0.113)	−0.001	0.105 (0.112)	0.011	0.110 (0.120)
	Time(post)	−0.143	0.095 (0.091)	−0.145	0.095 (0.091)	−0.100	0.098 (0.097)
	Intvn*time	−0.090	0.135 (0.148)	−0.088	0.135 (0.149)	−0.090	0.138 (0.155)
	prop.score			0.647	0.384 (0.379)	0.743	0.407 (0.403)
	male gender					0.217	0.069 (0.068)
	age = 18					0.703	0.089 (0.086)
	age = 19-20					1.079	0.088 (0.086)
	α	0.012		0.011		0.014	
Design-based	Intvn	0.005	0.109 (0.113)	−0.002	0.109 (0.112)	0.010	0.115 (0.121)
	Time(post)	−0.140	0.103 (0.091)	−0.143	0.103 (0.092)	−0.096	0.108 (0.098)
	Intvn*time	−0.093	0.146 (0.148)	−0.091	0.147 (0.149)	−0.094	0.154 (0.155)
	prop.score			0.660	0.384 (0.379)	0.761	0.408 (0.404)
	male gender					0.217	0.069 (0.068)
	age = 18					0.705	0.089 (0.086)
	age = 19-20					1.081	0.088 (0.085)
	α_0	0.017		0.016		0.021	
	α_1	0.007		0.006		0.007	

Estimates of the state-specific intercepts are not shown.

4. SAMPLE SIZE DETERMINATION

The general approach to power calculations in community trials is based upon an analysis of community summary statistics according to the study design. This follows from the fact that the intervention effect should be tested against the between-community variance. Since communities are assumed to be statistically independent, deriving an expression for the variance of the community summary statistic in each condition (intervention and control) is the crucial step in determining sample size. In this section we derive sample size formulae for the post-test only design, the pre-test–post-test nested cross-sectional design and the pre-test–post-test nested cohort design. Sample size formula for the nested cross-sectional design is applied to the national evaluation.

4.1. A general method based upon summary statistics

When the design involves communities followed under one of two conditions at one or two time points, the comparison of intervention and control communities is operationalized as a

one degree of freedom contrast. The general set-up is to specify the hypothesis of interest as $H_0: \delta = 0$ versus $H_1: \delta \neq 0$ where $\delta = \mu_1 - \mu_2$ is the intervention effect and $\mu_h = E(S_{hi})$, $h = 1, 2$ is the expected value of the summary statistic, S_{hi} , for intervention ($h = 1$) and control ($h = 2$) communities. The special task is to determine the variance of the summary statistic, $\sigma_h^2 = \text{var}(S_{hi})$, $h = 1, 2$. Then for a sufficiently large number of communities, the sample size needed for testing the intervention with possibly unequal numbers of communities per group using a two-sided test with α significance level and power $1 - \beta$ is

$$n_1 = \frac{(\sigma_1^2 + \sigma_2^2/r)(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2} \quad \text{and} \quad n_2 = \frac{(r\sigma_1^2 + \sigma_2^2)(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2} \quad (3)$$

where n_1 is the number of intervention communities and n_2 is the number of control communities, $r = n_2/n_1$, and z_c is the $(100 \times c)$ th percentile of the standard normal distribution [32]. Further, let $\Phi(\cdot)$ define its cumulative distribution function. For the purpose of sample size determination we assume $m = m_{it}$, for all i and t , where m_{it} is number of subjects in community i at time t . Our general approach is to define a statistic, $d = \sum_{i=1}^{n_1} S_{1i}/n_1 - \sum_{i=1}^{n_2} S_{2i}/n_2$, which is an unbiased estimator of δ . Power $(1 - \beta)$ is then given by

$$1 - \beta = \Phi\left(\frac{\delta}{\sqrt{\{\text{var}(d)\}}} - z_{1-\alpha/2}\right) \quad (4)$$

where $\text{var}(d) = \sigma_1^2/n_1 + \sigma_2^2/n_2$.

4.2. Sample size formula for post-test-only cross-sectional design

For the post-test-only nested cross-sectional design, the intervention effect, δ , is the comparison of the average response of subjects in intervention communities to the average response of subjects in control communities. The relevant summary statistic of interest for the i th community receiving the h th condition is the observed mean at post-test, $S_{hi} = \sum_{j=1}^m Y_{ij1}/m$. It is well established that for the post-test-only nested cross-sectional design the variance of the summary statistic is $\sigma_h^2 = \text{var}(S_{hi}) = \phi v_{ij1}/m$ where $\phi = 1 + (m - 1)\alpha_0$. In words, the variance of the summary statistic for a community in a given condition is a function of the variance of the youth-level outcome at post-test and the intraclass correlation among observations within a community [13]. In the case of continuous outcomes, under equal allocation ($n = n_1 = n_2 = K/2$) and equal variance ($\sigma_1^2 = \sigma_2^2$), expression (3) reduces to the sample size formula for comparison of means given in Donner and Klar (reference [13], p. 57).

For the purposes of sample size considerations, it is convenient in the remainder of the paper to replace the ij subscript in π_{ijt} and v_{ijt} with a single subscript h , because the only covariates in model equation (2) denote the time, $t = 0, 1$, and whether the community received the control ($h = 1$) or intervention ($h = 2$) condition. For binary data $\mu_1 = \pi_{11}$ and $\mu_2 = \pi_{21}$. The intervention effect, $\delta = \pi_{11} - \pi_{21}$, is the comparison of the proportion of subjects in intervention communities with the outcome to the corresponding proportion of subjects sampled in control communities in the post-test period. It follows that the variance of S_{hi} for binary outcomes, under $H_1: \delta \neq 0$, is

$$\sigma_h^2 = \frac{1}{m} \{[\pi_{h1}(1 - \pi_{h1})][1 + (m - 1)\alpha_0]\} \quad (5)$$

Under equal allocation, expression (3) reduces to

$$n = \frac{\phi\{(\pi_{11}(1 - \pi_{11}) + \pi_{21}(1 - \pi_{21}))\}(z_{1-\alpha/2} + z_{1-\beta})^2}{m(\pi_{11} - \pi_{21})^2} \quad (6)$$

which is equivalent to equation (5.5) in Donner and Klar (reference [13], p. 62), and the sample size formula in Section 4.2 of Liu and Liang [33]. Expression (6) is applicable to the model $g(\pi_{hij1}) = \beta_0 + \beta_1 x_{1i}$ for any standard link function because the hypothesis $H_0: \delta = 0$ is equivalent to $H_0: \beta_1 = 0$.

4.3. Sample size formula for pre-test–post-test cross-sectional design

For the pre-test–post-test nested cross-sectional design, the relevant contrast depends upon the link function. For model equation (2) with identity link function, $g(\pi_{ht}) = \pi_{ht}$, the test $H_0: \beta_{12} = 0$ corresponds to the contrast $\delta_m = \mu_1 - \mu_2 = (\pi_{11} - \pi_{10}) - (\pi_{21} - \pi_{20})$. The intervention effect, $\delta_m = \beta_{12}$, is the comparison of the average change over time of the intervention communities to the average change over time of the control communities. The relevant summary statistic of interest for the i th community is the change in the observed means from baseline to follow-up periods, $S_{hi} = \sum_{j=1}^m Y_{ij1}/m - \sum_{j=1}^m Y_{ij0}/m$. From first principles, its variance, given by $\sigma_h^2 = \text{var}(S_{hi})$, for the nested cross-sectional design is

$$\begin{aligned} \sigma_h^2 &= \frac{1}{m^2} \left\{ \text{var} \left(\sum_j Y_{ij1} \right) + \text{var} \left(\sum_k Y_{ik0} \right) - 2 \text{cov} \left(\sum_{j=1}^m Y_{ij1}, \sum_{k=1}^m Y_{ik0} \right) \right\} \\ &= \frac{1}{m} \{ (v_{h1} + v_{h0})\phi - 2m\alpha_1 \sqrt{(v_{h1}v_{h0})} \} \end{aligned} \quad (7)$$

In words, the variance of the summary statistic for a community in a given condition is a function of the variances of the youth-level outcome at each time point and the two types of intraclass correlation among observations within a community defined in the previous section.

When $n_1 = n_2$ and $v = v_{h1} = v_{h0}$, $h = 1, 2$, (a common assumption for continuous outcomes)

$$\text{var}(d) = \frac{4v}{nm} \{ 1 + (m - 1)\alpha_0 - m\alpha_1 \} \quad (8)$$

which is a special case of formula (9) of Feldman and McKinlay [8]. Expression (8) may be obtained with a reparameterization of the linear mixed model (their expression (1)) parameters to the population-averaged model parameters given by $v = \sigma_c^2 + \sigma_E^2$, $\alpha_0 = \sigma_c^2/v$ and $\alpha_1 = \rho_c \alpha_0$, where σ_c^2 , σ_E^2 , and ρ_c are the total cluster effects variance, residual variance and cluster autocorrelation as defined in their model. Note that σ_S^2 in their formula (9) is absorbed into σ_E^2 when there is only one observation per individual as considered here.

Following from (7), the variance of S_{hi} for binary outcomes, under $H_1: \delta_m \neq 0$, is

$$\sigma_h^2 = \frac{1}{m} [\{ \pi_{h1}(1 - \pi_{h1}) + \pi_{h0}(1 - \pi_{h0}) \} \phi - 2m\alpha_1 \sqrt{\{ \pi_{h1}(1 - \pi_{h1}) \pi_{h0}(1 - \pi_{h0}) \}}] \quad (9)$$

with values of π_{h0} and π_{h1} assigned uniquely for intervention and control conditions so that generally $\sigma_1^2 \neq \sigma_2^2$ in expression (3).

If inference is to be based upon the logistic model (model (2) with logit link), then the contrast of interest is $\delta_l = \text{logit}(\pi_{11}) - \text{logit}(\pi_{10}) - \text{logit}(\pi_{21}) + \text{logit}(\pi_{20})$. Generally, $H_0: \delta_m = 0$ and $H_0: \delta_l = 0$ are not equivalent hypotheses. Sample size determination may be based upon the summary statistic, $S_{hi} = \text{logit}(\sum_{j=1}^m Y_{ij1}/m) - \text{logit}(\sum_{j=1}^m Y_{ij0}/m)$, leading to $\sum_{i=1}^{n_h} S_{hi}/n_h$, an approximate unbiased estimator of $\mu_h = \text{logit}(\pi_{h1}) - \text{logit}(\pi_{h0})$. The approximate variance of the summary statistic is

$$\sigma_h^2 \approx \frac{1}{m} \left\{ \frac{\phi}{\pi_{h1}(1 - \pi_{h1})} + \frac{\phi}{\pi_{h0}(1 - \pi_{h0})} - \frac{2m\alpha_1}{\sqrt{\{\pi_{h1}(1 - \pi_{h1})\pi_{h0}(1 - \pi_{h0})\}}} \right\} \quad (10)$$

For $\alpha_0 = \alpha_1 = 0$, (10) is the asymptotic variance of the sum of two logits constructed from time-specific independent samples each of which involves m independent Bernoulli trials [34]. Note that both expressions (9) and (10) depend upon values of the four probabilities π_{ht} that define each contrast. Often researchers are unwilling or unable to specify particular values of the π_{ht} . Then values of π_{ht} may be chosen to give conservative values of σ_h^2 in (9) or (10) accordingly.

4.4. Sample size formula for pre-test–post-test cohort design

The hypotheses, contrast of interest (δ_m), and summary statistic are defined as for the pre-test–post-test cross-sectional design but now there is an additional correlation, α_2 , defined in Section 3.1, to take into account in determining σ_h^2 . Using first principles

$$\sigma_h^2 = \frac{1}{m} [(v_{h1} + v_{h0})\phi - 2\{\alpha_2 + (m - 1)\alpha_1\}\sqrt{(v_{h1}v_{h0})}] \quad (11)$$

When $n_1 = n_2$ and $v = v_{h1} = v_{h0}$, $h = 1, 2$, the variance of the estimated intervention effect is

$$\text{var}(d) = \frac{4v}{nm} [1 - \alpha_2 + (m - 1)(\alpha_0 - \alpha_1)] \quad (12)$$

Expressions (11) and (12) reduce to (7) and (8), respectively, when $\alpha_1 = \alpha_2$. Typically, $\alpha_1 < \alpha_2$, so the nested cohort design is more efficient than the nested cross-sectional design, barring drop-out.

4.5. Power analysis for the national evaluation

Many cluster-unit trials enrol a small number of clusters, each with a relatively large number of individuals, due to the often high expense of applying an intervention to each additional cluster, be it a community, school, medical practice, etc. However, in many government- and foundation-sponsored programmes, the resources available for the evaluation of the programme may be distinct from funds for implementing the intervention. Such was the case with the national evaluation. The total of 61 communities receiving intervention funds placed a restriction on possible study designs for the evaluation of the programme. Since seven potential intervention communities were not matched by the propensity score algorithm, only study designs with $n_1 \leq 54$ intervention communities were considered. Because the costs of implementing the evaluation stemmed in large part from the total number of surveys conducted $[(n_1 + n_2) \times 2m = N]$ and not so much to the number of communities evaluated, a decision was made to consider designs with $n_1 = 54$, $n_2 \geq 54$ and $N \approx 3680$. Note that possible values of n_2 were almost unlimited (Table I, $n_2 \leq 489$). The power of a potential design was determined by

Table V. Statistical power for comparing changes in average proportions for pre-test–post-test nested cross-sectional community designs (m, r) with $n_1 = 54$ intervention communities, $r = n_2/n_1$ control communities, and $2m$ individuals per community.

δ_m	$\rho = (0.018, 0)$				$\rho = (0.018, 0.009)$			
	(17,1.0)	(15,1.3)	(13,1.6)	(11,2.1)	(17,1.0)	(15,1.3)	(13,1.6)	(11,2.1)
0.15	0.988	0.988	0.987	0.983	0.994	0.994	0.992	0.989
0.14	0.975	0.976	0.973	0.966	0.986	0.986	0.983	0.977
0.13	0.952	0.954	0.950	0.940	0.971	0.971	0.966	0.956
0.12	0.916	0.919	0.913	0.900	0.945	0.945	0.937	0.923
0.11	0.863	0.867	0.860	0.842	0.902	0.902	0.892	0.872
0.10	0.791	0.796	0.787	0.768	0.839	0.839	0.826	0.803
0.09	0.701	0.707	0.698	0.677	0.754	0.755	0.741	0.715
0.08	0.597	0.604	0.595	0.574	0.652	0.653	0.638	0.612
0.07	0.487	0.493	0.485	0.467	0.537	0.538	0.524	0.500
0.06	0.378	0.383	0.376	0.362	0.419	0.421	0.409	0.389
0.05	0.278	0.282	0.278	0.267	0.309	0.310	0.302	0.287

For all entries, $\pi_{10} = \pi_{20} = \pi_{21} = 0.40$, and $\delta_m = (\pi_{11} - \pi_{10}) - (\pi_{21} - \pi_{20})$.

first solving for σ_1^2 and σ_2^2 in expression (9) by specifying $m, \alpha_0, \alpha_1, \pi_{10}, \pi_{11}, \pi_{20}$ and π_{21} . For the implied $\delta_m, n_1 = 54$, and a specified r , power $(1 - \beta)$ was determined for self-reported past 30-day use by solving expression (4). Although this was a non-randomized study, power calculations assumed $\pi_{10} = \pi_{20}$ since it was thought that the use of the propensity score method for selecting controls would result in the baseline outcomes being approximately balanced. Additionally, we assumed $\pi_{20} = \pi_{21} = 0.40$. We used the published value, $\alpha_0 = 0.018$, found in Table II of Murray and Short [19]; values of α_1 were not available so we considered two cases $\alpha_1 = 0$ and $\alpha_1 = \alpha_0/2$, with the latter being perhaps more realistic. We conservatively use Murray and Short's upper bound estimate of α_0 . Our goal was to identify a design (r and m) that maximized, or nearly maximized, statistical power with the restriction $n_1 = 54$. Four designs (r, m) were considered where each design has a total of approximately 1840 surveys for each time point. Of these, two designs, $n_1 = n_2 = 54, m = 17$ and $n_1 = 54, n_2 = 69, m = 15$, gave the highest power (Table V). The equal allocation ($r = 1$) design was chosen for the national evaluation. Note that the designs with the most clusters ($n_2 = 87$ and $n_2 = 113$) provide the least amount of power among those considered.

Next, we evaluated the sensitivity of statistical power to the intraclass correlations, $\rho = (\alpha_0, \alpha_1)$. Four sets of correlation parameter values were used, corresponding to no correlation, $\rho = (0, 0)$, 'low' correlation, $\rho = (0.006, 0.003)$, 'moderate' correlation, $\rho = (0.018, 0)$, and 'high' correlation, $\rho = (0.036, 0)$. The first is not a realistic assumption but provides a benchmark for assessing the overall effect of intraclass correlation on the power analysis. The low correlation is based upon Murray and Short's unadjusted estimate of α_0 while assuming $\alpha_1 = \alpha_0/2$. The moderate and high correlations are the upper bound and twice the upper bound reported by those authors. We conservatively use their unadjusted correlations in all instances. Under these assumptions, the minimum detectable difference (MDD) was defined as the smallest δ_m that the planned sample design could detect as statistically significant with a 0.05 level two-sided test and 80 per cent power. For $m = 17$, the MDD for δ_m with respect to changes in the proportions of past 30-day use were estimated to be 0.090, 0.092, 0.101 and 0.112 for

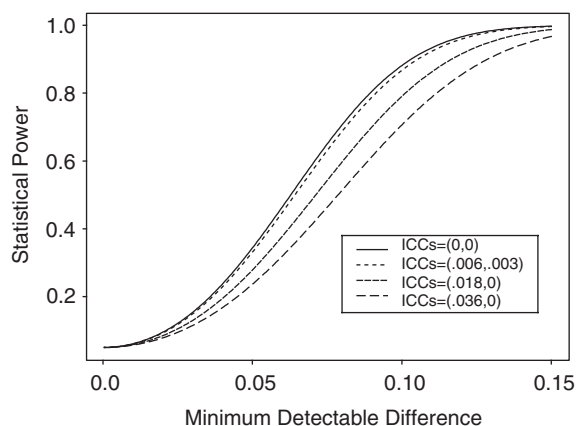


Figure 1. Power for past 30-day alcohol use, $n = 54$, $m = 17$.

the no, low, moderate and high correlation assumptions, respectively. Figure 1 presents power curves for each of the four sets of correlation values for past 30-day use for the design with $m = 17$. Power calculations were also carried out under a logit link and using δ_l and (10) in (4). The resulting MDD for δ_l for the same values of ρ considered above were 0.383, 0.399, 0.437 and 0.485, respectively. Assuming pre-test equivalence ($\beta_1 = 0$), these correspond to a range of odds ratios for the intervention effect of 1.47 to 1.62.

5. DISCUSSION

This paper presented sample size formulae for some common two-group cluster unit trial designs. Our sample size formulae are special cases of a general procedure for sample size determination based upon Wald tests in the GEE setting when the GEE model-based variance estimator is used [22]. Relative to general power methods for GEE [22, 33] that could be used if adjustment for covariates or propensity scores are desired, our sample size formulae are easy to use for the situations presented. In particular, special emphasis was given to formulae for the pre-test–post-test nested cross-sectional and cohort designs that are expressed in terms of population averaged model parameters. These formulae extend the class of correlation structures for which explicit sample size formulae for GEE Wald tests have been given [35]. Application of the proposed approach, for example, to a different contrast or study design, would require determination of the variance of the summary statistic of interest. Additionally, our formulae are likely to require modification for application to situations with small numbers of clusters. One possibility may be application of the t -distribution such as in the mixed model setting [36]. The utility of any sample size formula for cluster unit trials depends upon the availability of published estimates of intraclass correlations for various outcomes in the literature. There is a need for more publications presenting these values when reporting trial results.

This paper illustrated the use of propensity scores to obtain a comparable control group in a large non-randomized community trial. These methods are useful in the evaluation of

hundreds of government- and foundation-sponsored programmes for which the 'gold standard' of randomization is unobtainable. Their use depends upon the availability of information on baseline predictors of the outcomes of interest thought to be related to the selection of intervention communities. Although only four covariates were used in the design of the national evaluation, propensity scores may handle several variables since they are a scalar summary of them. Reducing several possible blocking factors to a single dimension is a general strategy among the various forms of restricted randomization sometimes used in group randomized trials [37]. While these procedures attempt to balance potential confounding covariates across study conditions, propensity scores apply to non-randomized trials in the selection of a comparable control group for a pre-existing intervention group.

There are some limitations in the propensity score method. In the national evaluation, these arose, in part, because the method was applied to each of the states separately and so the pool of potential control selections was limited. A caliper equal to 0.05 did not always give the number of matches desired and so additional matches were made using alternative criteria. As noted by one referee, selection criteria other than those based upon defining a caliper could be considered within the propensity score framework. One such design alternative might be to minimize the sum of squared differences in propensity scores between intervention and control communities. Whatever criterion is used in managing potential selection bias with propensity score matching, there is a trade-off in bias and generalizability. Use of a more liberal criterion (for example, selective use of a wider caliper to include communities with large populations in the national evaluation) may contribute to bias to the extent that covariates not balanced across study conditions are related to the study outcomes. The estimator of intervention effect may also be biased if important information on baseline predictors of the outcomes of interest used in the selection of intervention communities are not available either for use in propensity scores or through covariance adjustment. To ameliorate this threat, a pre-treatment period should be incorporated into the design for much the same reason that pre-test–post-test designs are recommended when there are a small number of communities in a randomized design. That is, when either the number of communities is small or randomization to conditions is not possible, there is no guarantee that communities in the two conditions are similar, and so comparisons should be made on the basis of within-community change. The use of propensity scores in selecting control communities in a pre-test–post-test design gives further strength to the argument that observed differences in mean changes between groups under different conditions are due to the intervention. Additionally, their use reduces the extent of extrapolation that would otherwise be required for inference based upon models with baseline covariates, since the differences among selected intervention and control communities are lessened. These considerations support a model-based approach in which inference is directed to the population of communities represented by communities in the sample with like covariates.

Various approaches are used in the analysis of cluster-unit trials with binary outcomes. There are three broad categories: analyses based upon population averaged models, subject-specific models, and summary statistics. The choice between population-averaged and subject-specific models, which can each handle cluster-level and individual-level covariates, should be based upon the scientific question of interest. Our view is that population-averaged models provide natural interpretations for many community intervention trials. While the generalized linear mixed model is often used for the analysis of many community trials [7, 11, 36, 38], there are important differences in the analysis of binary data between the two general approaches.

The generalized linear mixed model (with logit link) tends to give intervention parameter estimates that are larger than their population-averaged model counterparts [39–41]. Additionally, the mixed logistic model gives a treatment effect whose meaning is conditional upon the random effects and so applies only to that community, or, strictly speaking, to the population of communities with the same values of the unobserved random effect. On the other hand, the population-averaged model parameter that defines the intervention effect applies to the whole population. Aside from the linear model and a few other special cases, there are no simple relationships between variance components from generalized linear mixed models and marginal correlations defined for the distinct types of pairs of observations in a community [42]. Another compelling reason to consider a GEE approach is that for certain models (for example, expression (2)), it is asymptotically equivalent to an analysis based upon summary statistics. This follows from the equivalency of the corresponding power methods for Wald tests in these two approaches. More empirical work is needed to compare the various approaches in practical situations [44].

This paper applied GEE to a community trial with over a hundred clusters. The national evaluation is just one of many government- and foundation-sponsored programmes that enrol a moderate to large number of communities for which GEE may apply [1]. The fact that similar values of the standard error of the intervention effect were obtained for all three correlation structures considered in Table IV is consistent with the expectation that modelling correlation may not help improve efficiency of the estimator of the intervention effect for a study of this size. The primary benefit of applying GEE with design-based correlation structure to studies as large as the national evaluation may be to obtain correlation estimates for planning future studies. However, even in studies as large as the national evaluation, use of the model-based estimator with a design-based correlation structure may be a better choice than the sandwich estimator since the latter may be a relatively inefficient estimate of the true variance [17]. The fact that the validity of the model-based approach requires that R_i be correctly specified should not be considered a serious drawback when one considers that the competing approach based upon the generalized linear mixed model requires the full likelihood to be correctly specified.

The applicability of GEE methodology to cluster-unit trials with a small number of communities requires further research. The finite sample performance of GEE in the analysis of community intervention trials has been investigated for both discrete and continuous outcomes [11, 43]. In computer simulation studies of trials with 50 clusters or less, the performance of GEE in terms of estimated bias and test size of the intervention effect parameter was found to be less than desirable with the least favourable results occurring when the number of clusters was smallest [11, 43]. The poor test size results are partly due to the poor finite sample properties of the empirical sandwich variance estimator [14]. A recent simulation study of clustered binary data motivated by dental applications found that a GEE bias-corrected variance estimator ‘gave tests with sizes close to the nominal level even when the number of [clusters] was 10 and cluster sizes were unequal’, whereas the empirical sandwich variance estimator gave tests with sizes that could be two to three times the nominal level [14]. Other small sample adjustments in Wald-type tests are based upon estimating the degrees of freedom of approximate t - or F -tests [45, 46]. Further investigation is needed to explore the performance of these methods in situations common among community trials, specifically, those situations characterized by a small number of communities, each with a large number of subjects per community.

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